

Saumil Patel Machine Learning Engineer Beaumont, TX (936) 587-9020 saumilp9091@gmail.com LinkedIn GitHub Portfolio	
Summary: Senior Machine Learning Engineer with over 5 years of experience architecting and deploying AI/ML solutions across retail, medical device, and fintech industries. Proven expertise in developing end-to-end ML pipelines, computer vision systems, and deep learning models including LLMs and real-time object detection. Demonstrated success in sensor fusion, data engineering, and MLOps, consistently delivering measurable business impact through AI-driven solutions. Skilled in GPU optimization, parallel computing, and cloud platforms (AWS, Azure, GCP) with a strong focus on production-grade deployments and cross-functional collaboration.	
Skills: Programming languages: Libraries: Web Development: Probability and Statistics: Cloud and Databases: Machine Learning: Deep Learning: Analytics and Visualization: MLOps Tools: Hardware &, Sensors: LLMs & NLP: MLOps & Infrastructure: Python, JavaScript, MATLAB, R, SQL. TensorFlow, Keras, Scikit-Learn, PyTorch, Pandas, NumPy, Flask API, OpenCV, OpenAI Gym, JSON, TypeScript, JQuery, Bootstrap. HTML5, CSS, Node.js, React.js, CodePen.NET Core, ASP.Net, VB, WebAPI/Web Service, SQL Server, VS Code, Git, GitHub. Gaussian mixture models, hidden Markov model, Baye’s Theorem. AWS (EC2, S3, SageMaker), Google Cloud Platform, Microsoft Azure, Docker, Kubernetes, Redshift, MySQL, PostgreSQL, Google Colab, Oracle, ETL Spark, ETL Tools, CodePen, MS Visual Studio, Microsoft Office. Linear and Logistic Regression, Clustering, SVM, PCA, Random Forest, Boosting, Lasso, Ridge, CNN, RNN. TensorFlow, Keras. Jupyter Notebook, MS Excel, Tableau, Google Analytics. Weights & Biases, DVC, Git LFS, Docker Registry, Prometheus, Grafana Camera Systems, LiDAR, Radar, RFID, Edge Deployment LangChain, Transformers, BERT, GPT, Prompt Engineering Kubernetes, CI/CD, Git, DVC, MLflow, Kubeflow, Apache Airflow, Jenkins	
Experience: Lamar University - Beaumont, TX AI Researcher June 2023 - Dec 2024 Developed novel deep learning architecture for multi-sensor fusion architecture combining camera, LiDAR, and RFID data streams, achieving 98.9% accuracy in complex retail environments across 1538 product categories. Conducted extensive experiments with real-time object detection algorithms, optimizing performance to achieve 45-55 FPS on edge devices while maintaining high accuracy under varying lighting conditions. Researched and developed new approaches to automated drift detection in computer vision systems, reducing false positives by 75% through innovative vector similarity techniques. Investigated scalable approaches to real-time data processing, successfully handling 1TB+ daily data streams while maintaining sub-second latency for critical vision tasks. Pioneered new methodologies for multi-camera calibration using Aruco markers, reducing spatial alignment errors by 60% compared to traditional techniques. Medtronic - Santa Rosa, CA Sr. ML Engineer Oct 2022 - June 2023 Developed advanced medical imaging analysis system using deep learning, reducing inspection errors by 40% and saving \$1.8 million annually in quality control costs. Led implementation of automated MLOps pipeline with continuous monitoring, improving model deployment efficiency by 30% and reducing system downtime by 45%. Established comprehensive testing and validation framework, achieving 95% test coverage and reducing critical production issues by 60%. Designed real-time model performance monitoring system, enabling proactive maintenance and improving system reliability by 35%. Mentored team of 4 engineers in ML best practices, resulting in 25% faster project delivery and successful deployment of 3 major ML systems. Stryker - Mahwah, NJ ML Engineer Jun 2020 - Sep 2022 Architected end-to-end ML pipeline using AWS SageMaker, optimizing scheduling processes and reducing operational costs by \$2.1 million annually. Developed and deployed NLP-based customer service automation system, improving response accuracy by 30% and reducing manual processing time by 65%. Implemented containerized ML infrastructure on AWS, reducing deployment time by 40% and scaling to handle 3x increase in processing volume. Created automated quality monitoring system for ML models, reducing model degradation incidents by 75% and improving system reliability by 40%. Led adoption of MLOps best practices across team of 4 engineers, resulting in 35% improvement in up time and 50% reduction in production incidents. AEM Loan - San Francisco Bay Area Data Analyst Aug 2015 - Dec 2017 Managed loan processing workflow optimization, reducing application review time by 40% through process automation and team coordination. Led team of 8 employees implementing new loan origination system, achieving 25% improvement in processing efficiency. Developed business intelligence dashboards using Tableau, providing insights that increased loan approval rate by 30%. Facilitated agile workflow implementation between business and IT teams, reducing project delivery delays by 25%.	
Education: Lamar University, Beaumont TX Doctor of Engineer in Industrial and Systems Engineering August 2021 – December 2024 Lamar University, Beaumont TX Master of Engineering Science in Industrial and Systems Engineering January 2017 – August 2021 Northwestern Polytechnic University, Fremont, CA Master of business Administration in project management August 2014 – December 2015 Gujarat Technological University, Ahmedabad, India Bachelor of Engineering in Mechanical Engineering September 2009 – August 2013	